32. Consider the following statements regarding the earthquakes:

1. **The intensity of earthquake is measured on Mercalli Scale.**
2. **The magnitude of an earthquake is a measure of energy released.**
3. **Earthquake magnitudes are based on direct measurements of the amplitude of seismic waves.**
4. **In the Richter Scale, each whole number demonstrates a hundredfold increase in the amount of energy released.**

Which of these statements are correct:

(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1 and 4 (d) 1 and 3

Ans. (a) : On a ritcher scale each increase of one unit on the scale represents a Ten fold increase in the magnitude of an earthquake. Each increase of one unit also represent 31 times more energy than the previous whole number on the scale. Hence 4th statement is incorrect. Rest of the statements are correct.

33. Cloudy nights are warmer compared to clear cloudless nights, because clouds:

(a) Prevent cold waves from the sky from descending on earth
(b) Reflect back the heat given off by earth
(c) Produce heat and radiate it towards earth
(d) Absorb heat from the atmosphere and send it towards earth.

Ans. (b) : Cloudy nights are warmer compared to clear cloudless nights, because clouds Reflect back the heat given off by earth. At night cloud cover has the opposite effect. If skies are clear, heat emitted from the earth's surface freely escapes into space, resulting in colder temperatures. However, if clouds are present, some of the heat emitted from the earth's surface is trapped by the clouds and reemitted back towards the earth.

34. Which one of the following weather conditions is indicated by a sudden fall in barometer reading?

(a) Stormy weather
(b) Calm weather
(c) Cold & dry weather
(d) Hot & sunny weather

Ans. (a) : Sudden fall in barometer reading indicate stormy weather. Storms are created when a center of low pressure develops with a system of high pressure surrounding it. It may be marked by significant disruptions to normal conditions such as strong wind, tornados, hail, thunder and lightning.

35. If the stars are seen to rise perpendicular to the horizon by an observer, he is located on the:

(a) Equator (b) Tropic of Cancer
(c) South Pole (d) North Pole

Ans. (a) : If the stars are seen to rise perpendicular to the horizon by an observer then the observer is at the equator because celestial equator is an imaginary circle around the sky directly above the earth's equator. It is always 90º from the pole. All the stars rotate in a path that is parallel to the celestial equator.

36. The high density of population in Nile Valley and Island of Java is primarily due to:

(a) Intensive agriculture
(b) Industrialization
(c) Urbanization
(d) Topographic constraints

Ans. (a) : The Nile is the main river of Africa. It is known as the Nile River after the white and blue Nile rivers meet at Khatun in Sudan. The Nile River is called the boon of Egypt because of its fertile agricultural land characteristic of the Nile region. Most of Africa's population is inhabited in the areas around the Nile River.

The island of Java, which is close to the equator, also receives high temperatures as well as high rainfall. As a result, the agricultural land here is also fertile.

37. Consider the following statements made about the sedimentary rocks:

1. **Sedimentary rocks are formed at earth's surface by the hydrological system.**
2. **The formation of sedimentary rocks involves the weathering of pre existing rocks.**
3. **Sedimentary rocks contain fossils.**
4. **Sedimentary rocks typically occur in layers.**

Which of these statements are correct?

(a) 1 and 2 (b) 1 and 4
(c) 2, 3 and 4 (d) 1, 2, 3 and 4

Ans. (d) : Sedimentary rocks are formed by aggregation of sediments. Due to the formation of different layers of sediments, they are called flattened rocks. Sedimentary rocks are composed in a certain order.

They are formed by aggregation of rock powder, fauna and flora. Hence fossils are found in sedimentary rocks. Sedimentary rocks cover 75% of the Earth's surface.

Most of the mineral and petroleum products are found in these rocky areas.

38. Volcanic eruptions do not occur in the:

(a) Baltic Sea (b) Black Sea
(c) Caribbean Sea (d) Caspian Sea

Ans. (a) : Volcanic eruptions occur only in certain places and do not occur randomly. This is because the Earth's crust is broken into a series of slabs known as tectonic plates. Sixty percent of all active volcanoes occur at the boundaries between tectonic plates. Volcanic eruptions do not occur in the Baltic Sea.

39. A class of animals known as Marsupials is a characteristic feature of

(a) Africa (b) Australia
(c) South America (d) South-east Asia

Ans. (b) : Marsupials are a group of mammals commonly having pouches of skin where mothers murses their pre mature babies. These marsupians are found in Australia. An example of this is the kangaroo. Most kangaroos in the world are found in Australia.

40. The temperature and rainfall of a meteorological station are given below:

	Temperature ºC	Rainfall (cm)
J	9.4	12.2
F	10.6	9.1
M	11.7	7.9
A	12.2	2.5
M	13.3	1.8
J	13.9	0.3
J	13.9	–
A	14.4	–
S	15.6	0.8
O	15.0	2.5
N	13.3	6.1
D	10.6	11.7

Average Temperature : 12.8ºc
Average Rainfall : 54.9cm per annum

Identify the region having the above climatic pattern from amongst the following:

(a) Mediterranean region
(b) Monsoon region
(c) Steppe region
(d) N.W. European region

Ans. (a) : The above mentioned climatic pattern indicates the Mediterranean region with average temperature 12.8°C and average rainfall 59.9 cm.

41. Match List I with List II and select the correct answer using the code given below the lists:

List I (Local Wind)	List II (Region)
A. Fohn	1. Argentina
B. Simoom	2. Kurdistan
C. Santa Ana	3. California
D. Zonda	4. Alps

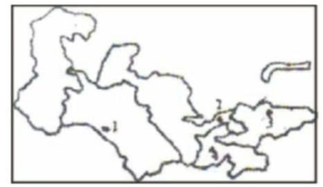
	A	B	C	D
(a)	2	4	1	3
(b)	4	2	3	1
(c)	2	4	3	1
(d)	4	2	1	3

Ans. (b) : Fohn-Alpes, Simoom-Kurdistan, Santana California, Zonda-Argentina, Chinook-USA, Siracoo-Italy, Mistral-France, Harmattan-Guinea Coast, Blizzard-Polar regions, Brickfilder – Australia , Berg wind - South Africa

42. The given maps shows four towns of Central Asian region marked as 1, 2, 3 and 4. Identify these from the following list and select the correct answer using the codes given below:

Town:

1. Bishkek 2. Ashgabat
3. Tashkent 4. Dushanbe



	A	B	C	D
(a)	3	1	2	4
(b)	3	1	4	2
(c)	1	3	2	4
(d)	1	3	4	2

Ans. (a) : Ashgabat is marked as '1' Tashkent is marked as '2' Bishkek is marked as '3' Dushanbe is marked as '4' in the given map.

43. Assertion (A) : During the Neap Tides, the high tide is lower and the low tide is higher than usual.

Reason (R) : The Neap Tide, unlike the Spring Tide, occurs on the New Moon instead of on the Full Moon.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (b) : The main reason for the origin of tides is the attraction power of the sun and the moon relative to the earth. Daily tides come once in 12 hours 26 minutes. The time between one tide and the next ebb is 6 hours 15 minutes.

When the sun, moon and earth are in a straight line, there is a state of great tide. In this case, the height of the tide is maximum. The maximum time for the tide to be at maximum height is 6 seconds.

However both (A) and (R) are individually correct, but R does not explains A.